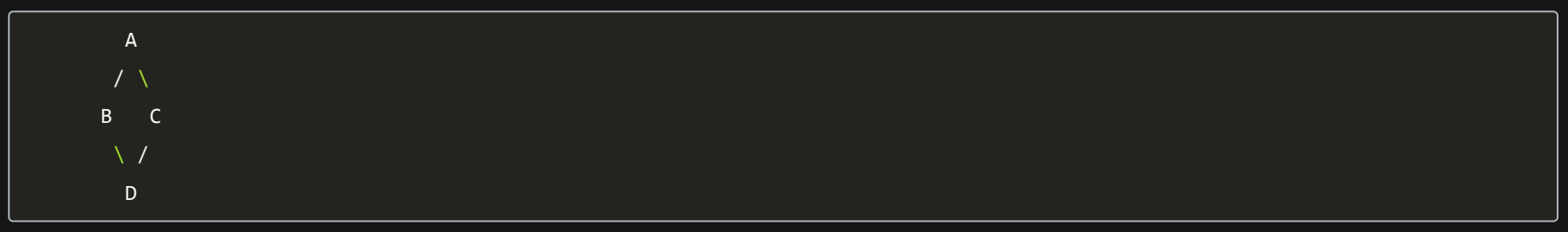
Diamond Problem in C++

This is a **very common C++ OOP interview question**.

1. What is the Diamond Problem?

Consider this inheritance structure:



- `B` inherits from `A`
- `C` inherits from `A`
- `D` inherits from both `B` and `C`

This forms a **diamond shape**.

Example

```cpp
class A {
public:
    int x = 10;
};

class B : public A {};
class C : public A {};

class D : public B, public C {};
```

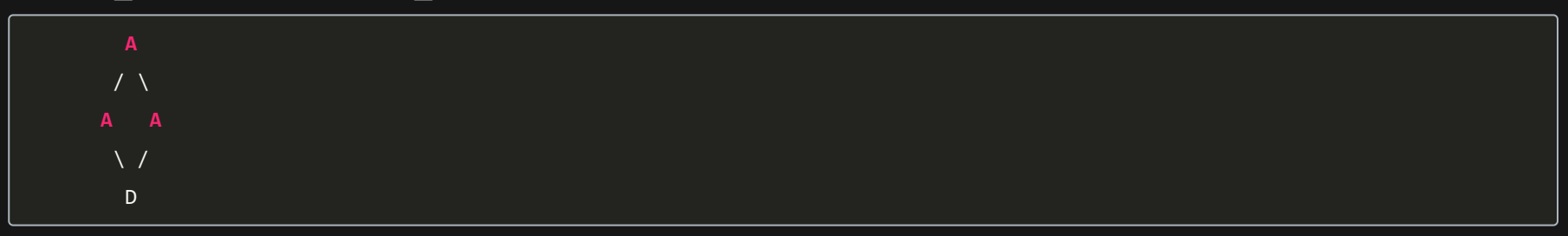
Now:

```cpp
D obj;
cout << obj.x;
```

❌ **Compilation error: ambiguous**

Why?

Because `D` contains **two copies of** `A`:



One `A` comes through `B`, and another comes through `C`.

So the compiler doesn't know whether:

```cpp
obj.x
```

means:

```cpp
obj.B::A::x
```

or:

```cpp
obj.C::A::x
```

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2. How can we resolve the ambiguity?

You could explicitly specify the path:

```cpp
D obj;

cout << obj.B::x;
cout << obj.C::x;
```

But this **doesn't solve the underlying problem**.

There are still **two separate** `A` **objects** inside `D`.

Usually, we want `D` to have only **one shared** `A`.

That's where **virtual inheritance** comes in.

3. Virtual Inheritance

We change:

```
class B : public A {};
class C : public A {};
```

to:

```
class B : virtual public A {};
class C : virtual public A {};
```

Full example:

```
class A {
public:
    int x = 10;
};

class B : virtual public A {};

class C : virtual public A {};

class D : public B, public C {};
```

Now the structure is conceptually:

```
    A
   / \
  B   C
   \ /
    D
```

But unlike normal inheritance, **B and C share the same `A` subobject**.

So `D` effectively has:

```
    A
   / \
  B   C
   \ /
    D
```

with **only one A**.

Therefore:

```
D obj;

cout << obj.x;  // 10
```

☑ No ambiguity.

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4. What does `virtual` actually mean here?

This is important:

```
class B : virtual public A
```

does **not** mean that `A`'s functions become virtual.

It means:

> **The base class `A` is inherited virtually, so the most-derived class shares a single `A` subobject among all inheritance paths.**

So:

```
class B : virtual public A {};
class C : virtual public A {};
```

means:

> "If someone eventually inherits from both B and C, don't create separate copies of A."

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5. Why is this useful?

Suppose `A` contains state:

```
class A {
public:
    int value;

    A(int v) : value(v) {}
};
```

Without virtual inheritance:

```
        A
       / \
      B   C
       \ /
        D

D contains:

A::value   ← through B
A::value   ← through C
```

Which `value` should `D` use?

With virtual inheritance:

```
        A
       / \
      B   C
       \ /
        D

D contains:

one A::value
```

So there is a **single shared base state**.

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6. Important: Constructor behavior

This is a good interview follow-up.

With virtual inheritance, **the most-derived class constructs the virtual base**.

```
class A {
public:
    A(int x) {
        cout << "A\n";
    }
};

class B : virtual public A {
public:
    B() : A(10) {}
};

class C : virtual public A {
public:
    C() : A(20) {}
};

class D : public B, public C {
public:
    D() : A(30) {}
};
```

When creating:

```
D obj;
```

`A(30)` is used.

Not `A(10)` or `A(20)`.

Interview line to remember

> **The most-derived class is responsible for constructing the virtual base class.**

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7. Normal vs Virtual Inheritance

Normal inheritanceVirtual inheritanceMultiple copies of common base possibleOnly one shared virtual baseDiamond can cause ambiguityResolves diamond ambiguity`B : public A``B : virtual public A`Each path has its own `A`Paths share the same `A`

🎯 Interview Questions

Q1. What is the diamond problem?

Answer:

> The diamond problem occurs in multiple inheritance when two intermediate classes inherit from the same base class and a derived class inherits from both intermediate classes. This can result in two copies of the common base class and cause ambiguity when accessing its members.

Q2. How does virtual inheritance solve the diamond problem?

Answer:

> Virtual inheritance ensures that only one shared instance of the common base class exists in the most-derived object, eliminating duplicate base subobjects and ambiguity.

Q3. Is this `virtual` the same as a virtual function?

No.

These are different concepts:

```
// Virtual function
virtual void foo();
```

is related to **runtime polymorphism / dynamic dispatch**.

While:

```
class B : virtual public A {};
```

is related to **virtual inheritance / shared base subobject**.

Q4. Does virtual inheritance mean there is no `A` object?

No.

There is still **one** `A` **subobject**. The difference is that `B` and `C` share it rather than each containing their own copy.

⭐ The 20-second interview answer

If the interviewer asks **"Explain the diamond problem and virtual inheritance"**, say:

> "The diamond problem occurs when a class inherits from two classes that both inherit from the same base class. Without virtual inheritance, the final class gets two copies of the common base, causing ambiguity. C++ solves this using virtual inheritance, where the intermediate classes virtually inherit the common base, ensuring that the most-derived class contains only one shared instance of that base. The most-derived class is also responsible for constructing that virtual base."

That is the answer I'd memorize for interviews.
